EYE MOVEMENTS, PUPILLARY REFLEXES & AUTONOMIC CONTROL OF VISION

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Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- ✔ Identify the intraocular & extraocular muscles involved in eye movements.
- ✔ Describe the major types of eye movements in horizontal, sagittal & anteroposterior planes.
- ✔ Identify the cranial nerves involved in voluntary eye movements.
- ✔ Describe the sympathetic & parasympathetic innervation of eye.
- ✔ Describe pupillary reflexes & their neural pathways.
- ✔ Define clinical conditions i.e. Argyll Robertson pupil, Horner syndrome & presbyopia.

INTRODUCTION

- Eyes are the sensory organs that allow us to see.
- Eyeballs sit in the bony cavities called ***the orbit***.
- Human eye ball is approximately globe-shaped with a diameter of about 24mm.
- Eyeball is made up of two segments, an ***anterior part*** & a ***posterior part***.
- Anterior part is small & forms one sixth of the eyeball.
- Posterior part is larger & forms five sixths of the eyeball.
- The front visible part of eye is made up of the whitish ***sclera***, a coloured ***iris*** & the ***pupil***.
- A thin layer called the ***conjunctiva*** sits on top of this.

OCULAR MUSCLES

Muscles of the eyeball are of two types.

I. Intrinsic muscles

II. Extrinsic muscles

INTRINSIC MUSCLES

- Intrinsic muscles are formed by smooth muscle fibers & are controlled by autonomic nerves.
- Intrinsic muscles of eye are:

1. Constrictor pupillae (iris sphincter)

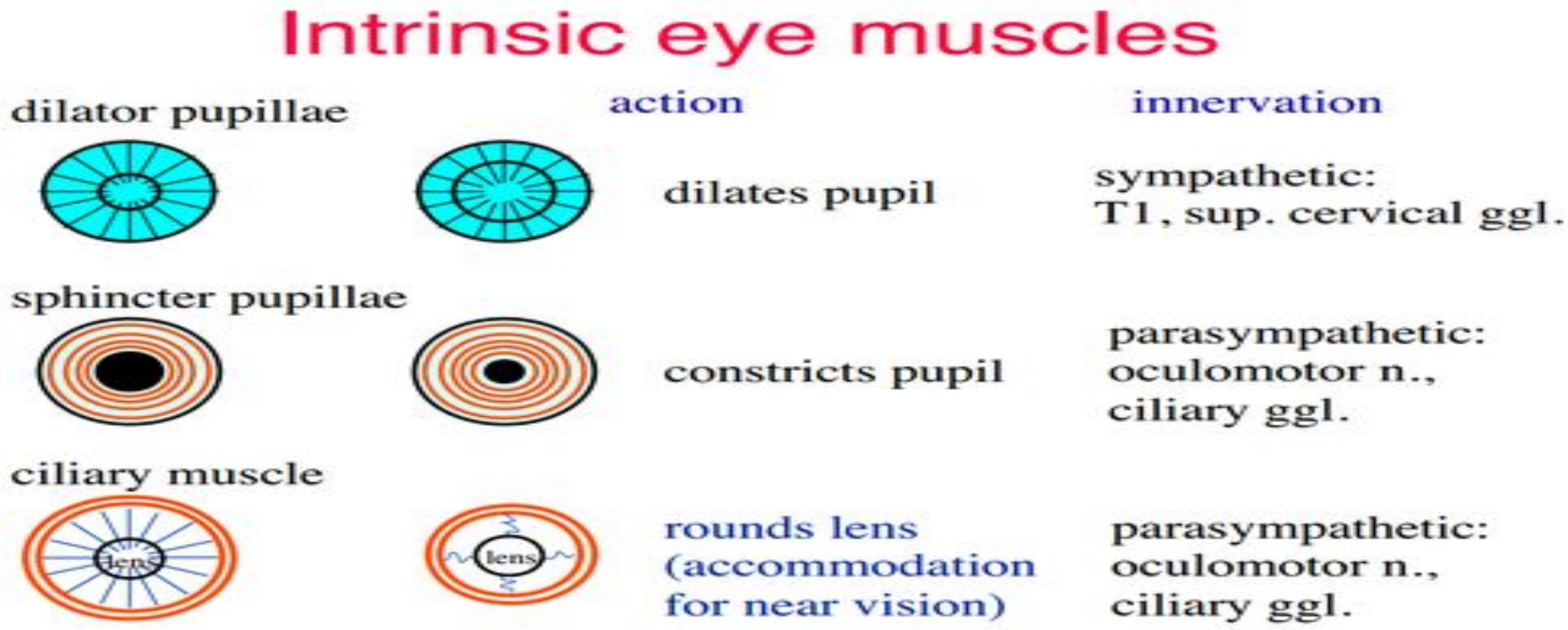
It is formed by circular muscle fibers & contraction of this muscle causes constriction of pupil.

2. Dilator pupillae

It is formed by radial muscle fibers & contraction of this muscle causes dilation of pupil.

3. Ciliary muscle

Contraction of ciliary muscle increases the anterior curvature of lens during accommodation.

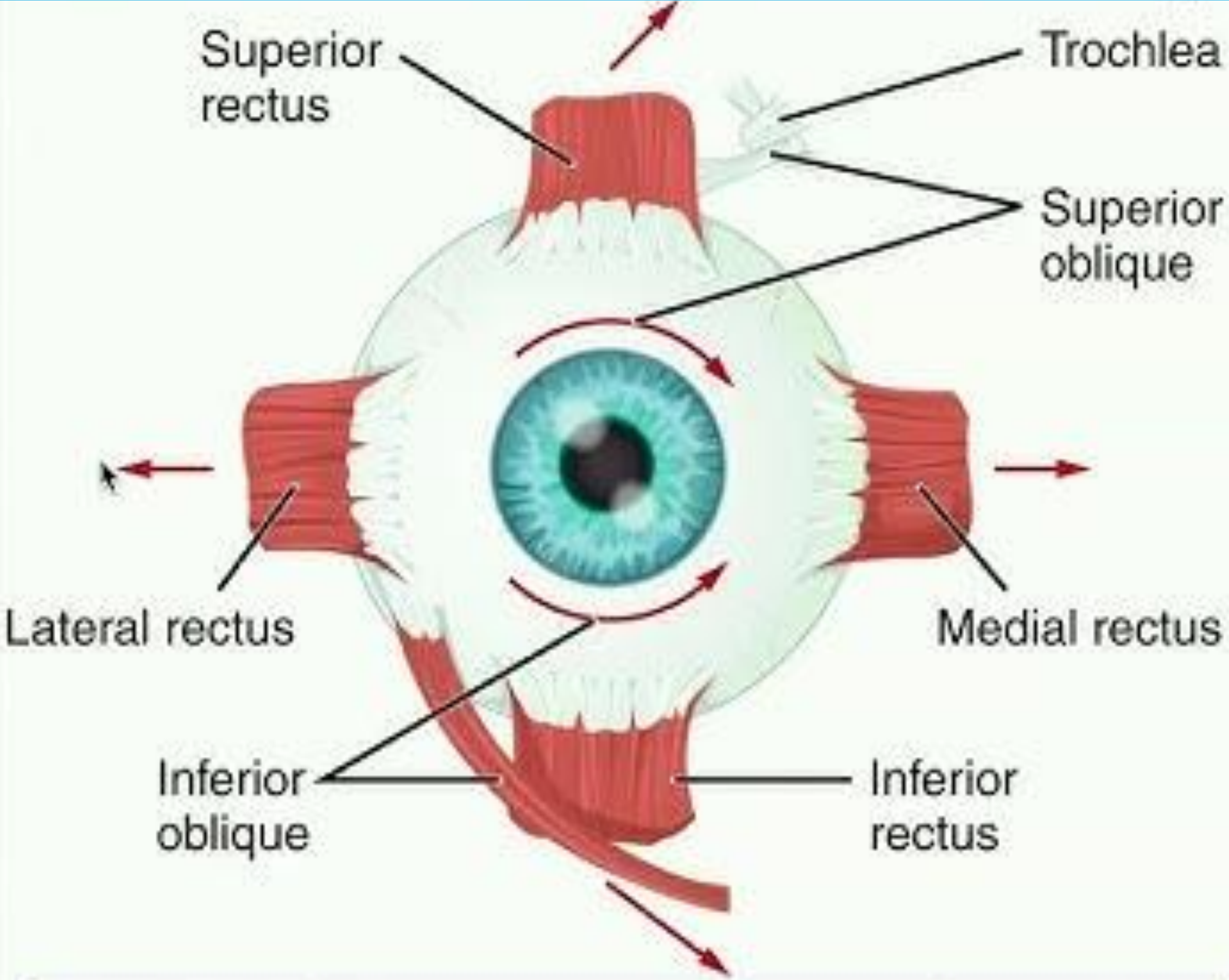


EXTRINSIC MUSCLES

Extrinsic muscles are formed by skeletal muscle fibers & are controlled by the somatic nerves.

- Eyeball moves within the orbit by six extrinsic skeletal muscles.
- One end of each muscle is attached to the eyeball & the other end to the wall of orbital cavity.
- There are four straight muscles(rectus) & two oblique muscles. Extrinsic muscles of the eyeball are:

1. Superior rectus
2. Inferior rectus
3. Medial or internal rectus
4. Lateral or external rectus
5. Superior oblique
6. Inferior oblique



Muscle	Action	Controlling cranial nerve
Lateral rectus	Moves eye laterally	VI (abducens)
Medial rectus	Moves eye medially	III (oculomotor)
Superior rectus	Elevates eye and turns it medially	III (oculomotor)
Inferior rectus	Depresses eye and turns it medially	III (oculomotor)
Inferior oblique	Elevates eye and turns it laterally	III (oculomotor)
Superior oblique	Depresses eye and turns it laterally	IV (trochlear)

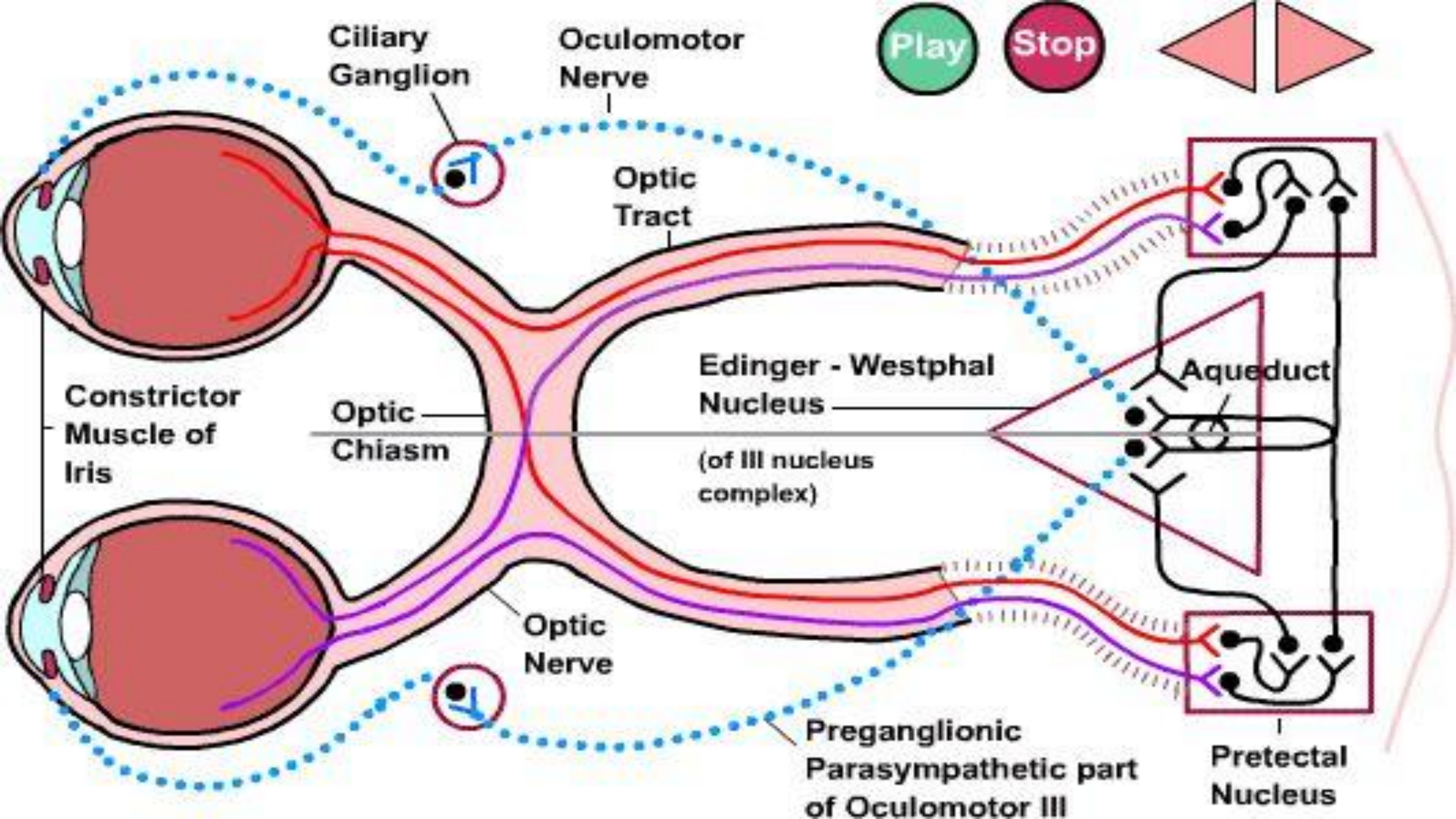
- Extrinsic eye muscles (6) are skeletal muscles that rotate the eye in its socket.
- Each is controlled by one of three cranial nerves (III, IV, and VI).

AUTONOMIC INNERVATION OF INTRINSIC MUSCLES

- Intrinsic muscles of eyeball are innervated by both sympathetic & parasympathetic divisions of autonomic nervous system.

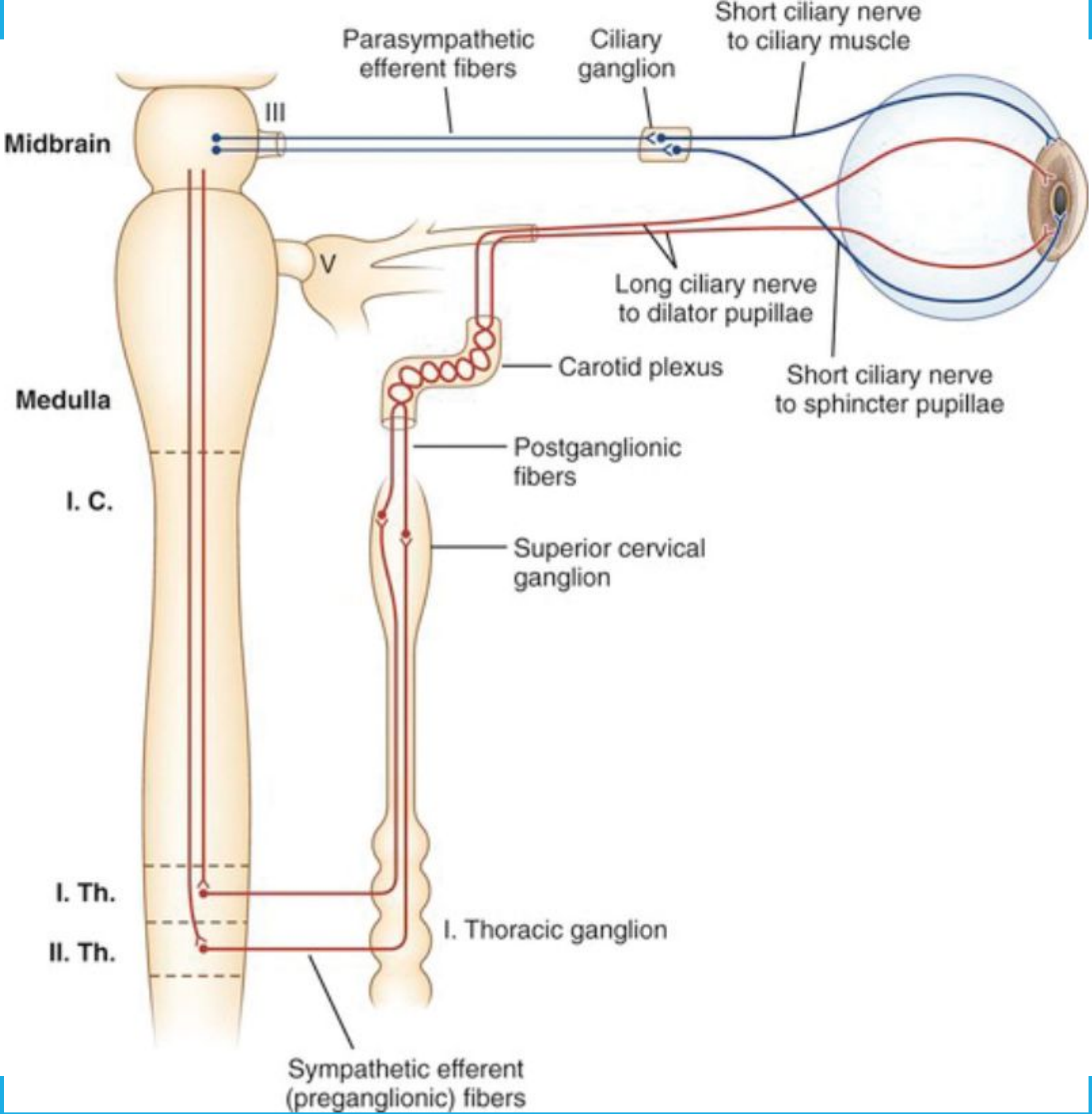
PARASYMPATHETIC INNERVATION

- Parasympathetic preganglionic fibers arise from Edinger-Westphal nucleus of cranial nerve lll.
- After passing through the cranial nerve lll, these fibers synapse with postganglionic fibers in the ciliary ganglion.
- Postganglionic fibers then pass through ciliary nerves & innervate the ciliary muscle & constrictor pupillae.
- Stimulation of parasympathetic nerve fibers causes contraction of ciliary muscle & constrictor pupillae.


Play
Stop
Ciliary Ganglion
Oculomotor Nerve
Optic Tract
Edinger - Westphal Nucleus
(of III nucleus complex)
Aqueduct
Constrictor Muscle of Iris
Optic Chiasm
Optic Nerve
Preganglionic Parasympathetic part of Oculomotor III
Pretectal Nucleus

SYMPATHETIC INNERVATION

- Sympathetic preganglionic nerve fibers arise from lateral horn of first thoracic segment of spinal cord.
- These fibers then pass through the sympathetic chain & synapse with the neurons of superior cervical sympathetic ganglion.
- Postganglionic fibers arising from this ganglion run along with carotid artery & its branches to reach the intrinsic muscles of the eyeball.
- Stimulation of sympathetic nerve fibers causes relaxation of ciliary muscle & contraction of dilator pupillae.


Parasympathetic efferent fibers
Ciliary ganglion
Short ciliary nerve to ciliary muscle
Midbrain
III
V
Long ciliary nerve to dilator pupillae
Carotid plexus
Short ciliary nerve to sphincter pupillae
Medulla
Postganglionic fibers
I. C.
Superior cervical ganglion
I. Th.
I. Thoracic ganglion
II. Th.
Sympathetic efferent (preganglionic) fibers

INNERVATION OF EXTRINSIC MUSCLES

- Extrinsic muscles of eyeball are innervated by somatic motor nerve fibers.
- Somatic nerve fibers arise from cranial nerve nuclei in the brainstem & reach the ocular muscles via three cranial nerves.

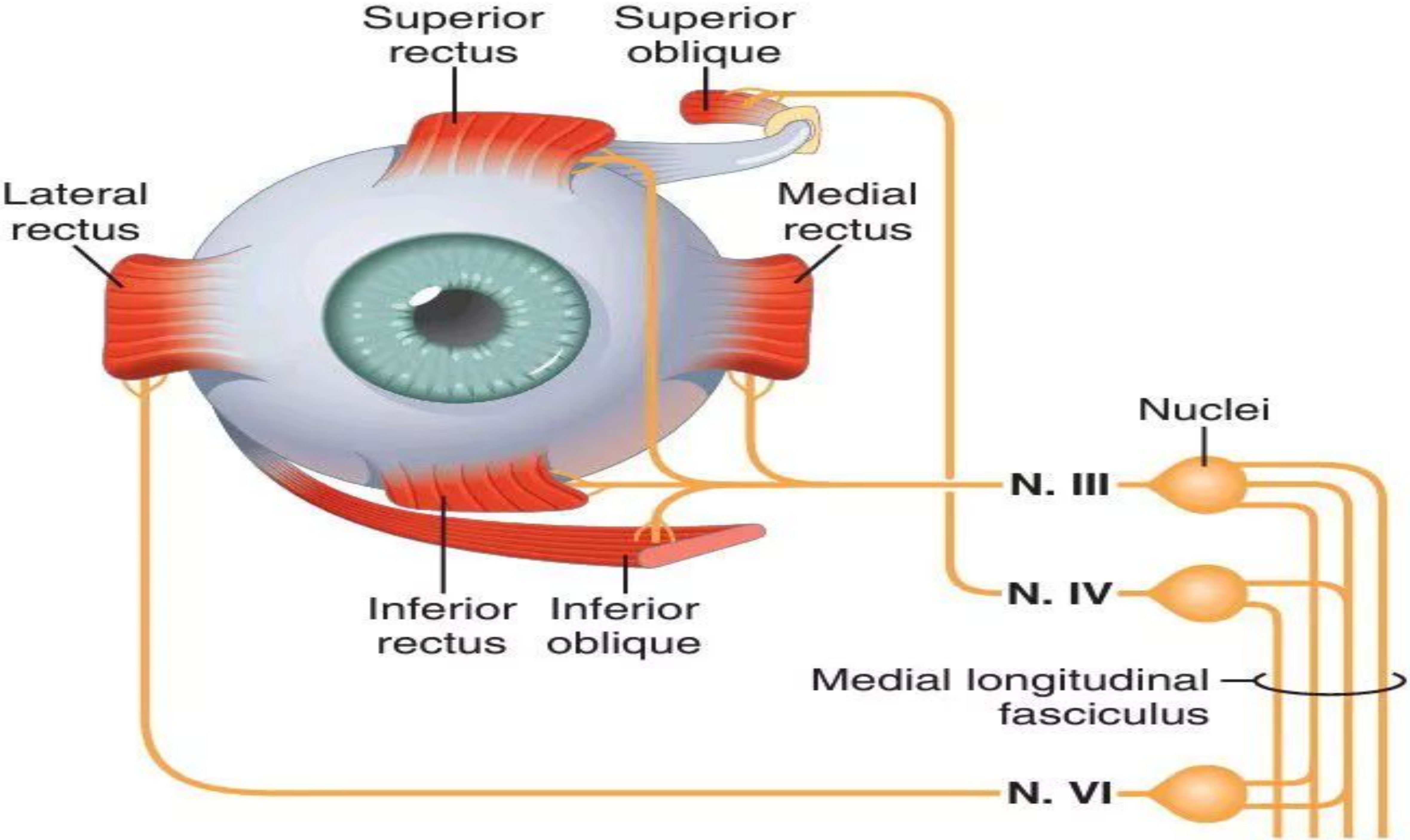
1) Oculomotor nerve(CN III)
2) Trochlear nerve(CN IV)
3) Abducens nerve(CN VI)

1. Oculomotor nerve

- Oculomotor nerve supplies the superior rectus, inferior rectus, medial rectus & inferior oblique muscles of the eyeball.

2. Trochlear nerve

- Trochlear nerve supplies the superior oblique muscles of the eyeball.


Superior rectus
Superior oblique
Lateral rectus
Medial rectus
Nuclei
N. III
N. IV
N. VI
Inferior rectus
Inferior oblique
Medial longitudinal fasciculus

3. Abducens nerve

Abducens nerve supplies the lateral rectus muscles.

OCULAR MOVEMENTS

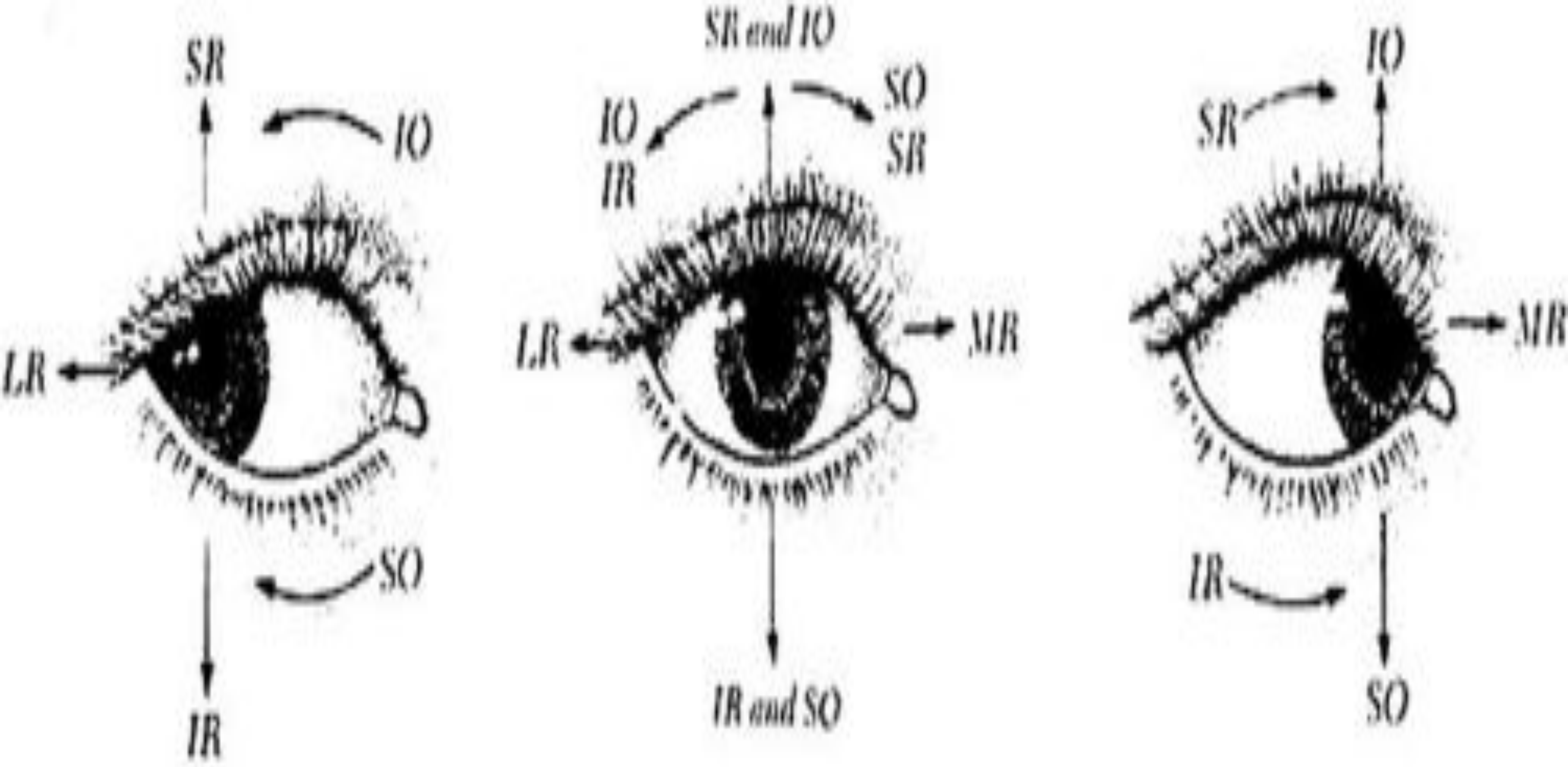
- Eyeball moves or rotates within the orbital socket in any of the three primary axes i.e. ***vertical***, ***transverse*** & ***anteroposterior*** axis.

MOVEMENTS IN VERTICAL AXIS

- Movements of eyeball in vertical axis or in horizontal plane are of two types:

I. ***Abduction or lateral movement(outward movement) ->*** occurs due to the contraction of ***lateral rectus*** mainly but supported by the two oblique muscles.

II. ***Adduction or medial movement(inward movement) ->*** occurs due to the action of ***medial rectus*** muscle along with the action of ***superior rectus*** & ***inferior rectus*** muscles.



Abduction Primary position Adduction

MOVEMENTS IN TRANSVERSE AXIS

- Movements of eyeball in transverse axis or in sagital plane are of two types:

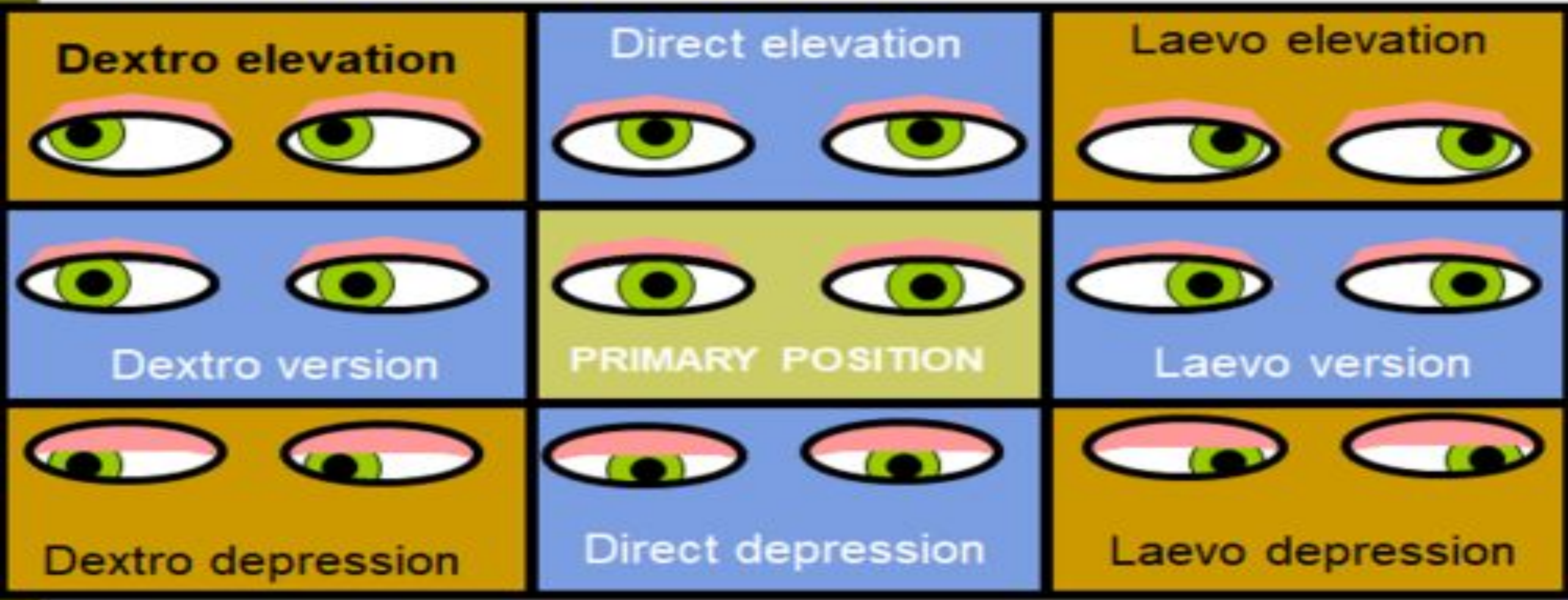
I. ***Elevation or upward movement ->*** occurs due to the contraction of ***superior rectus*** & ***inferior oblique*** muscles.

II. ***Depression or downward movement ->*** brought out by the ***inferior rectus*** & ***superior oblique*** muscles.

MOVEMENTS IN ANTEROPOSTERIOR AXIS

- Movements of eyeball in anteroposterior axis or in the frontal plane are called ***torsion or wheel movements***.
- Torsion movements are of two types namely ***extorsion*** & ***intorsion***.

Directions of versions

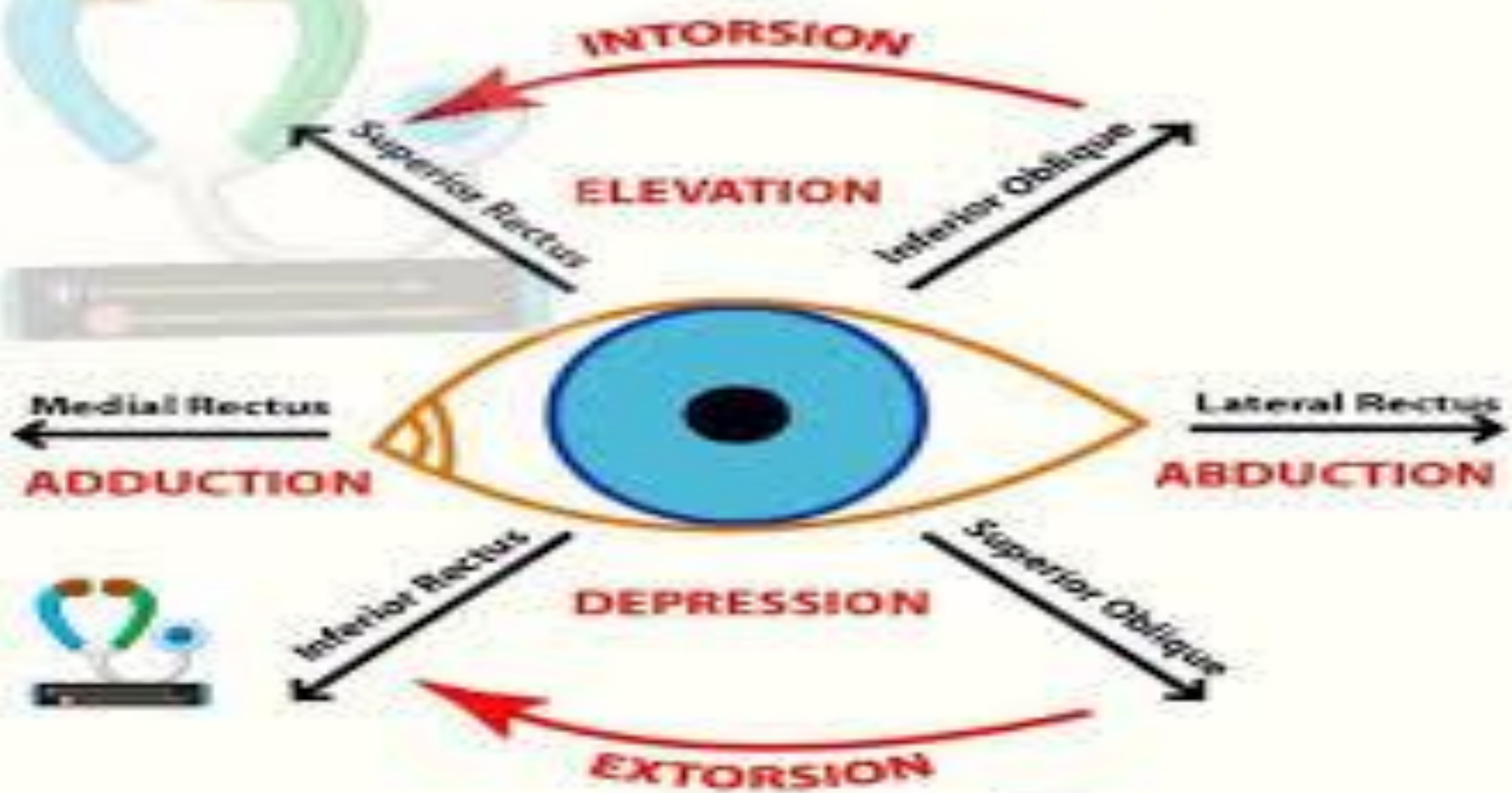


Dextro right **Laevo left**

CONT..

1. ***Extorsion ->*** during extorsion, the eyeball is rotated in such a way that the cornea turns in upward & outward direction & is due to the contraction of ***inferior oblique*** & ***inferior rectus*** muscles.
2. ***Intorsion ->*** during intorsion, eyeball is rotated in such a way that the cornea moves in downward & inward direction & is due to the contraction of ***superior oblique*** & ***superior rectus*** muscles.

EXTRAOCULAR MOVEMENTS



SIMULTANEOUS MOVEMENTS OF BOTH EYEBALLS

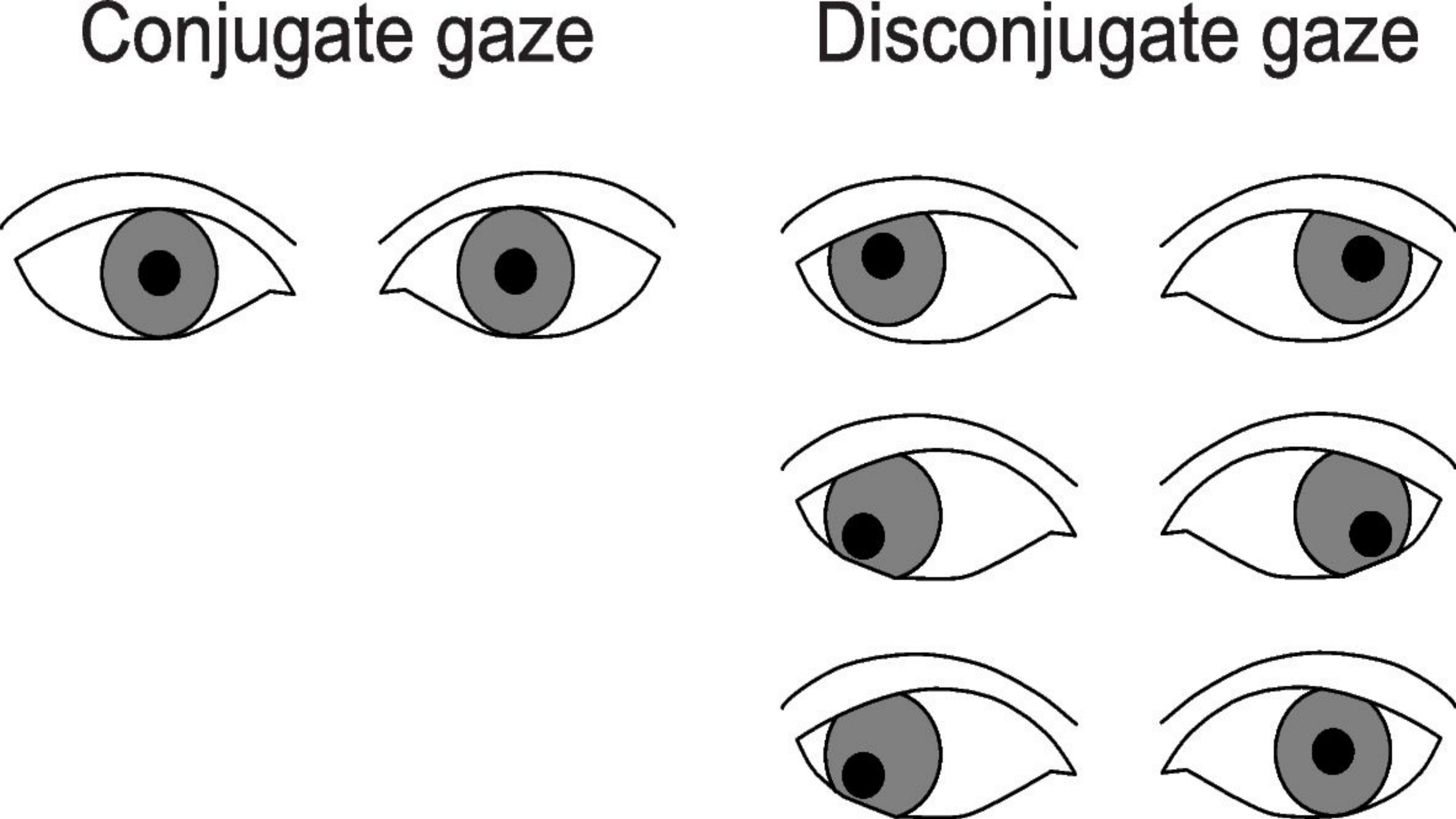
- Simultaneous movements of both eyeballs are of four types:

1. CONJUGATE MOVEMENT

- It is the movement of both eyeballs in same direction.
- Visual axis of both eyes remain parallel.
- This type of movement is due to the contraction of ***medial rectus of one eye*** & ***lateral rectus of the other eye***.

2. DISJUGATE MOVEMENT

- Disjugate movement is the movement of both eyeballs in opposite direction.
- Disjugate movement is of two types named as ***convergence*** & ***divergence***.


Conjugate gaze
Disconjugate gaze

Convergence:

It is the movement of both eyeballs towards nose & is due to the simultaneous ***contraction of medial rectus*** & simultaneous ***relaxation of lateral rectus*** of both eyes. Visual axes move close to each other. Convergence of eyeballs occur during accommodation.

Divergence:

It is the movement of both eyeballs towards temporal side & is due to the simultaneous ***contraction of lateral rectus*** & simultaneous ***relaxation of medial rectus*** of both eyes. Visual axes of eyes move away from each other.

3. PURSUIT MOVEMENT

- It is the movement of eyeballs along with the object when eyeballs follow a moving object.

4. SACCADIC MOVEMENT

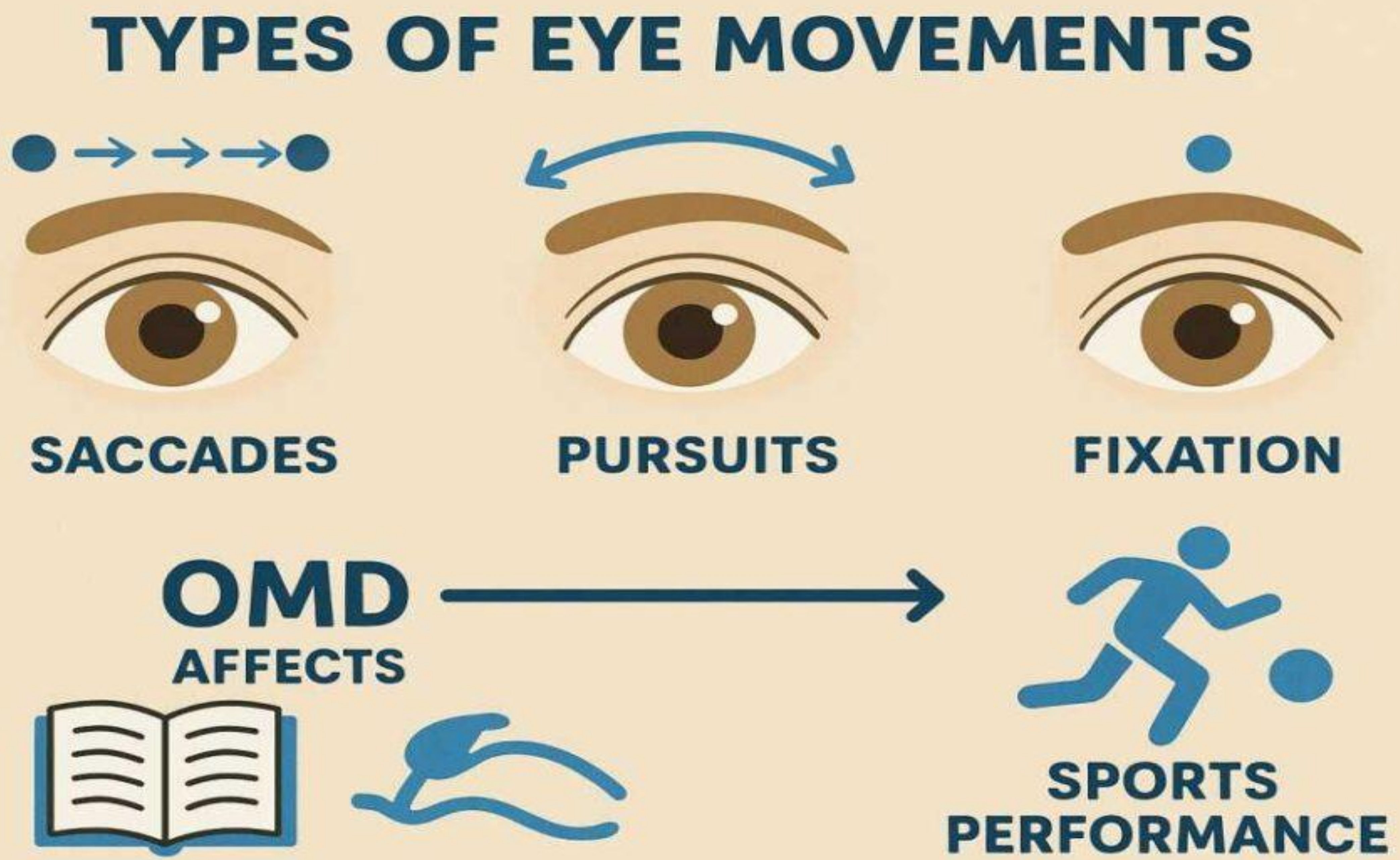
- Saccadic movement is the quick jerky movement of both eyeballs when the fixation of eyes(gaze) is shifted from one object to another. Saccadic movement is also called ***optokinetic movement***.

Convergence

When we look up close our eyes angle inwards towards our nose. This happens when we look at a book, cell phone, or do work at a desk.

Divergence

When we look in the distance our eyes are more centered or slightly angled outwards. This happens when looking across the room, when we look ahead while driving, and when reading signs.


TYPES OF EYE MOVEMENTS
SACCADES
PURSUITS
FIXATION
OMD
AFFECTS
SPORTS
PERFORMANCE

PUPILLARY REFLEXES

- Reflexes that alter the size of pupil are called pupillary reflexes.
- There are three types of pupillary reflexes namely:

1. Light reflex
2. Ciliospinal reflex
3. Accomodation reflex

1. LIGHT REFLEX

- In pupillary light reflex, pupil constricts when light is flashed into the eyes.

There are two types of light reflex:

I. Direct light reflex
II. Indirect light reflex

DIRECT LIGHT REFLEX

- Direct light reflex is the direct reaction of pupil to the light.
- In this reflex, pupil constricts in an eye when light is thrown into that eye.

INDIRECT LIGHT REFLEX

- In indirect light reflex also called ***consensual light reflex***, constriction of pupil occurs in both eyes when light is thrown into one eye.
- Constriction of pupil also occurs in the opposite eye even no light rays fall on that eye.

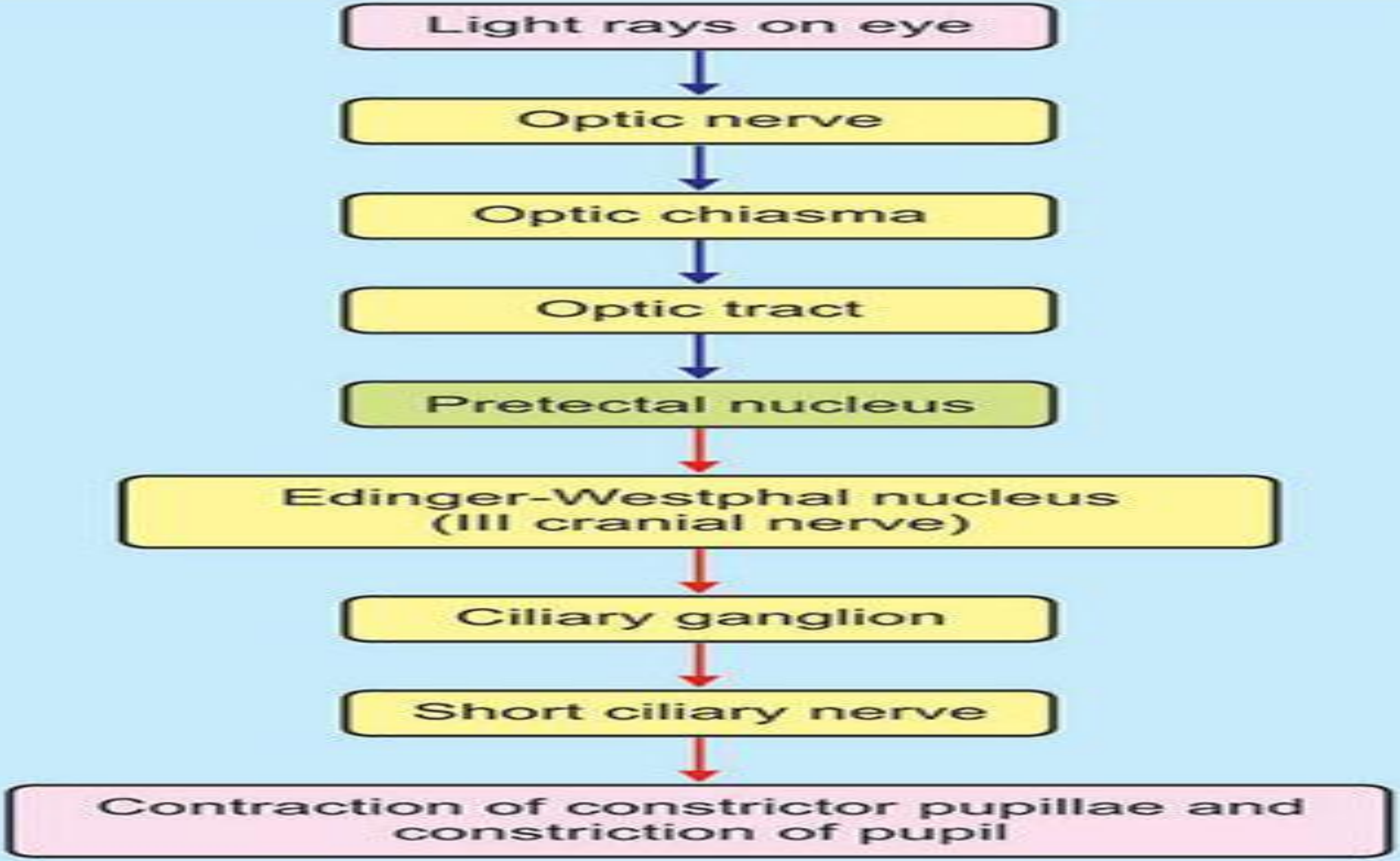
PATHWAY FOR LIGHT RFLEX

AFFERENT PATHWAY

- When light falls on the eye, the visual receptors are stimulated & afferent impulses from these receptors travel through the optic nerve, optic chiasma & optic tract.
- Fibers of light reflex pathway & visual pathway are same upto the optic tract.
- At midbrain level, few fibers get separated from optic tract & synapse with the neurons of pretectal nucleus.
- ***Pretectal nucleus*** of midbrain constitutes the ***center for light reflexes***.

EFFERENT PATHWAY

- Efferent impulses from pretectal nucleus are carried through short fibers to the rostral portion of Edinger-Westphal nucleus of the oculomotor nerve(CN III).
- From Edinger-Westphal nucleus, preganglionic fibers pass through the oculomotor nerve to reach the ciliary ganglion.
- Postganglionic fibers arising from the ciliary ganglion pass through the short ciliary nerves & reach the eyeball to cause contraction of ***constrictor pupillae muscle***.



Light rays on eye
Optic nerve
Optic chiasma
Optic tract
Pretectal nucleus
Edinger-Westphal nucleus (III cranial nerve)
Ciliary ganglion
Short ciliary nerve
Contraction of constrictor pupillae and constriction of pupil

2. CILIOSPINAL REFLEX

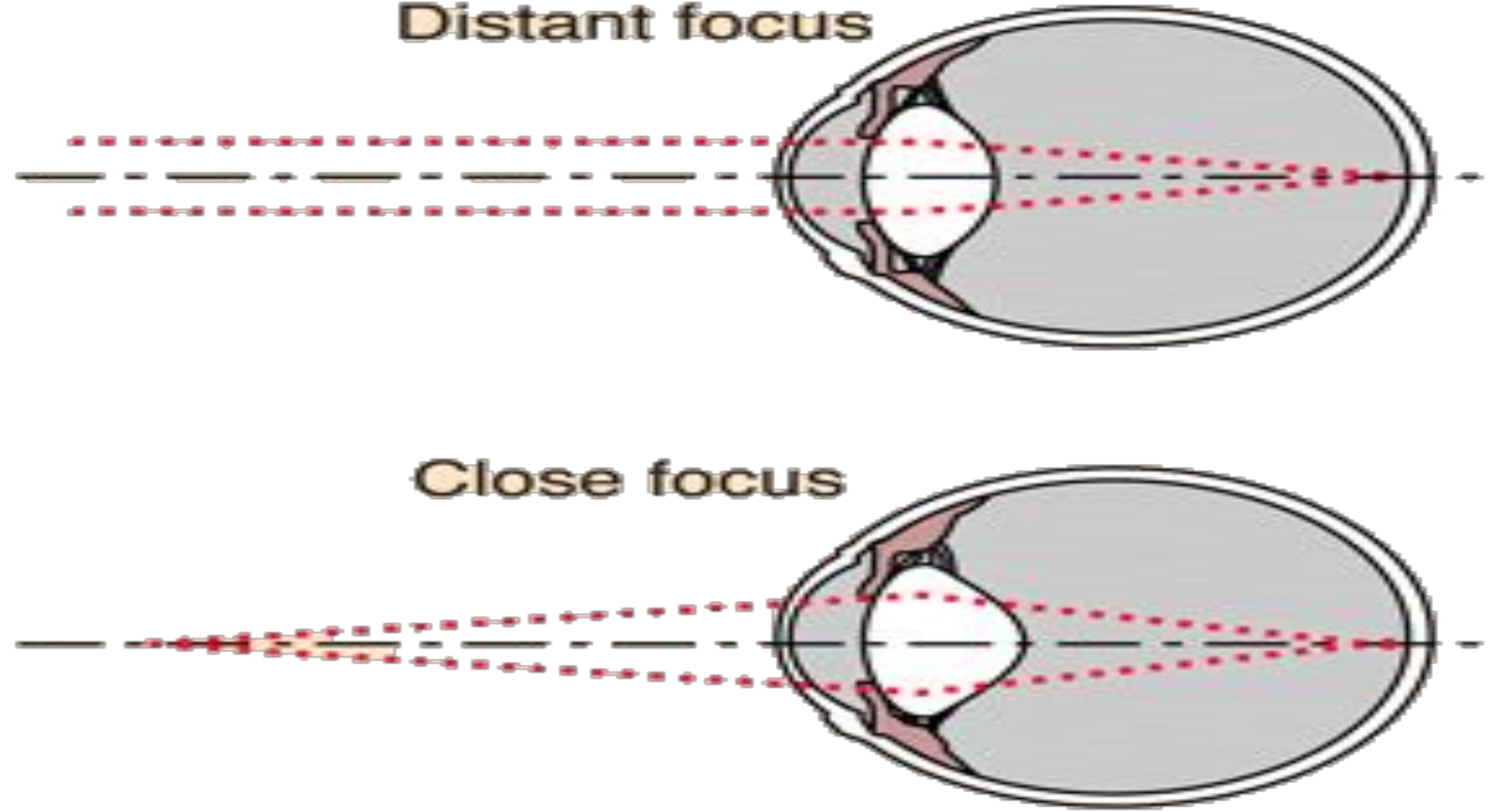
- Ciliospinal reflex causes dilation of pupil in the eyes due to the painful stimulation of skin over the neck.
- It is because of the contraction of dilator pupillae muscle.
- Sensory impulses travel through the cutaneous afferent nerve.
- The center for this reflex is in the first thoracic spinal segment.
- Efferent impulses pass through the sympathetic fibers & reach the dilator pupillae muscle.

3. ACCOMODATION REFLEX

- Accomodation is the adjustment of eye to see near or distant objects clearly.
- In accommodation, light rays from near or distant objects are brought to a focus on sensitive part of retina.
- Accomodation is achieved by various adjustments in the eyeball.

MECHANISM OF ACCOMODATION

- Light rays from distant objects are approximately parallel & less refracted when getting focused on retina whereas light rays from near objects are divergent & should be refracted(converged) to a greater extent to be focused on retina.
- Accomodation in humans is achieved by the alteration in the convexity of lens to alter the refractory power of lens as needed.



Distant focus
Close focus

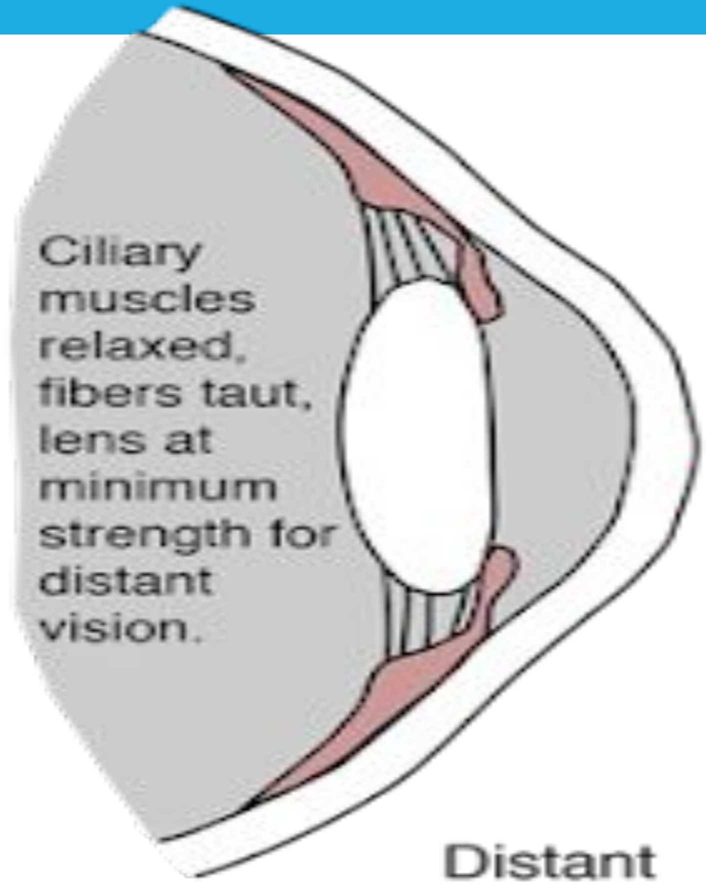
YOUNG-HELMHOLTZ THEORY

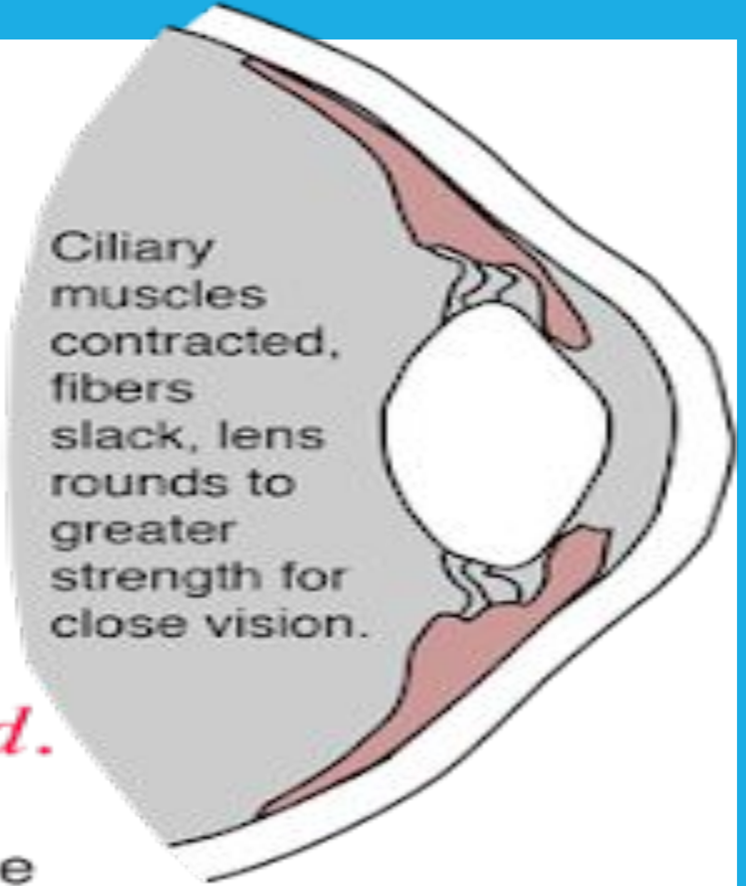
- The mechanism of accommodation in human eye was first suggested by ***Young*** & later supported by ***Helmholtz***.
- Young-Helmholtz theory explains how the curvature of lens increases to enhance the refractive power of the lens.
- While seeing a distant object(distant vision), lens is flat due to the traction of suspensory ligaments that extend from lens capsule & attached to ciliary processes.
- Ciliary processes are attached to choroid through the ciliary muscle.
- When vision is shifted from distant object to a near object(near vision),ciliary muscle contracts & draws the choroid forward which brought the ciliary processes closer to lens.

CONT..

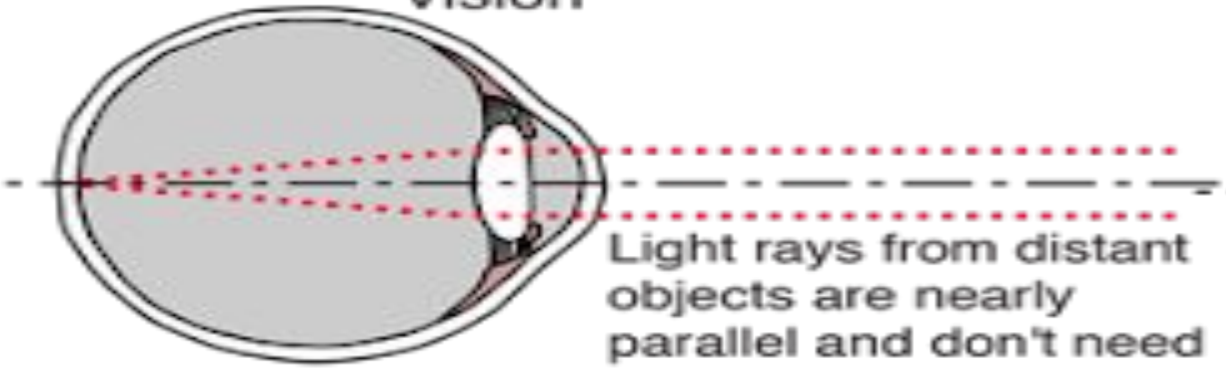
- Suspensory ligament slackened & the tension on the lens is released.
- Lens bulges forward due to its elastic property which increases the anterior curvature(convexity) of the lens.
- In resting eye, tension in choroids is maintained by the intraocular pressure that pulls the ciliary processes backward & outward that tenses up the suspensory ligament & the lens become flat.

The eye accommodates for close vision by tightening the ciliary muscles, allowing the pliable crystalline lens to become more rounded.



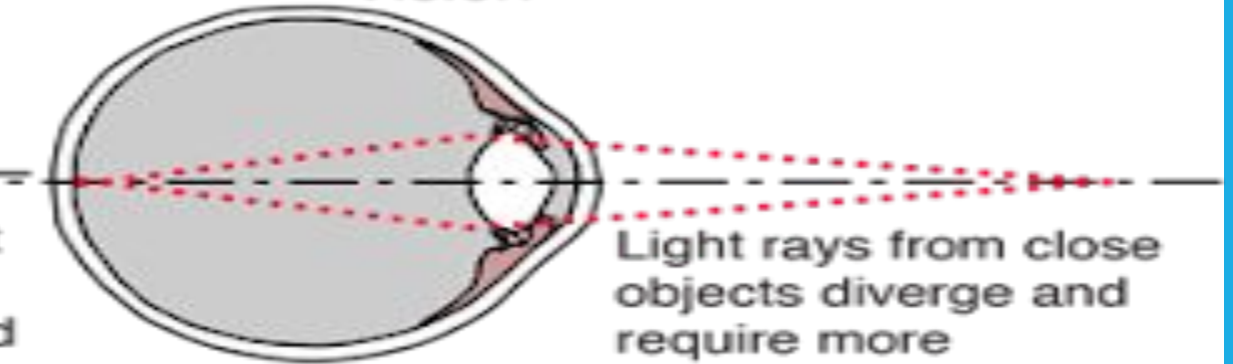


Distant Vision



Light rays from distant objects are nearly parallel and don't need as much refraction to bring them to a focus.

Close Vision



Light rays from close objects diverge and require more refraction for focusing.

PURKINJE-SANSON IMAGES

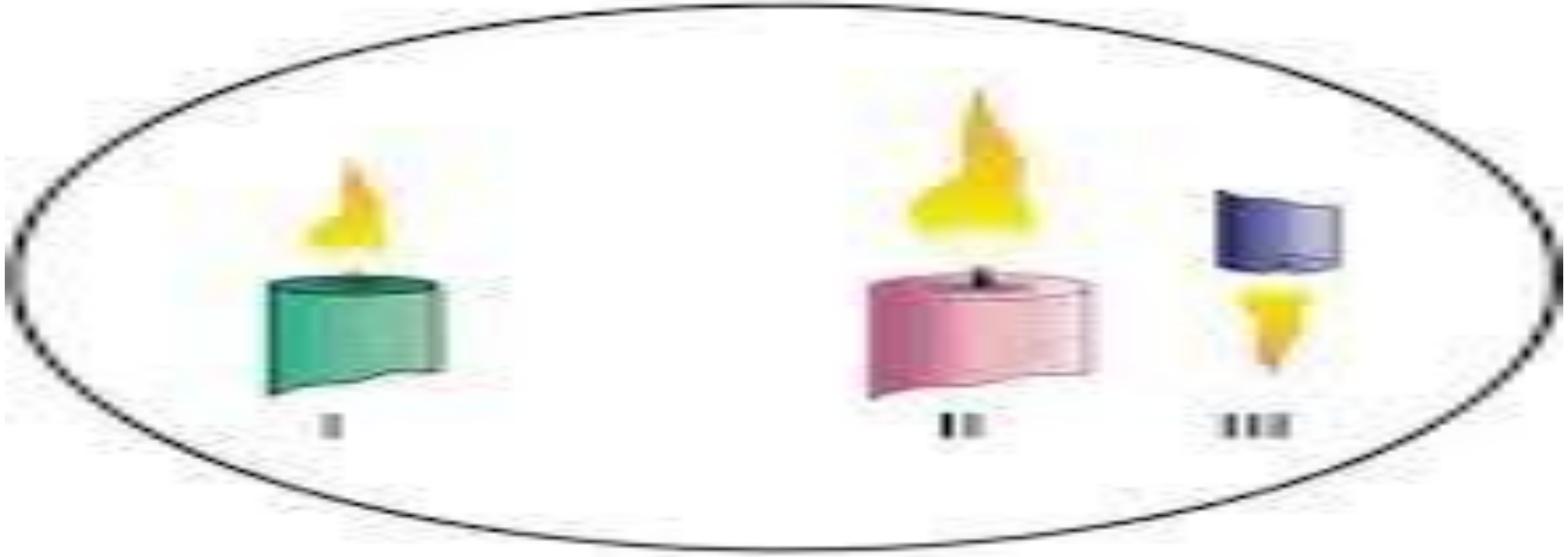
- Purkinje-sanson images are used to demonstrate the change in convexity of lens during the process of accommodation for near vision.

EXPERIMENT:

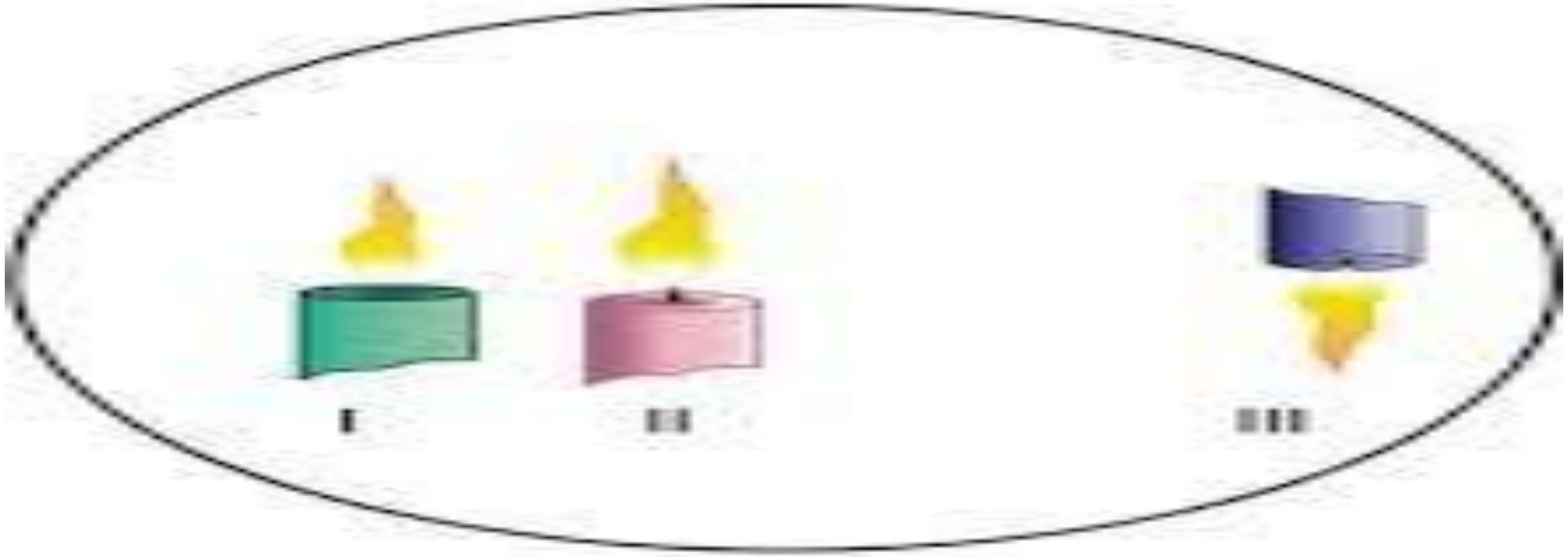
- Subject is made to sit in a darkroom & a lighted candle is held in front of the subject whose one eye is opened & the other eye is closed.
- Three images of the flame are seen in the opened eye:

1. First image is upright & bright. It shines from the surface of cornea acting as a mirror.
2. Second image is upright but dim. It is reflected from the anterior convex surface of the lens.
3. Third image is inverted & small. It is formed by the posterior surface of lens acting as a concave mirror.

During distant vision



During near vision



ACCOMODATION USING PURKINJE-SANSON IMAGES

- When the person looks at a distant object(distant vision), the second image reflected from the anterior surface of lens is near the third image formed from the posterior surface of the lens.
- During accommodation for near vision, no changes takes place in the first & third image but the second image becomes smaller & moves closer towards the first image.
- The change in size & position of the second image shows the increased convexity of anterior surface of lens during accommodation process for near vision.

OTHER ADJUSTMENTS IN THE EYEBALL DURING ACCOMODATION

- In addition to the increase in anterior curvature(convexity) of the lens, two other adjustments are also made in the eyeball during accommodation for near vision:

1. Convergence of both eyelids to bring the retinal images onto the corresponding points.
2. Constriction of pupil to increase the visual acuity(by reducing lateral chromatic & spherical aberrations), to reduce the amount of light entering the eye & to increase the depth of focus through more central part of lens as the convexity is increased.

PATHWAY FOR ACCOMODATION REFLEX

AFFERENT PATHWAY

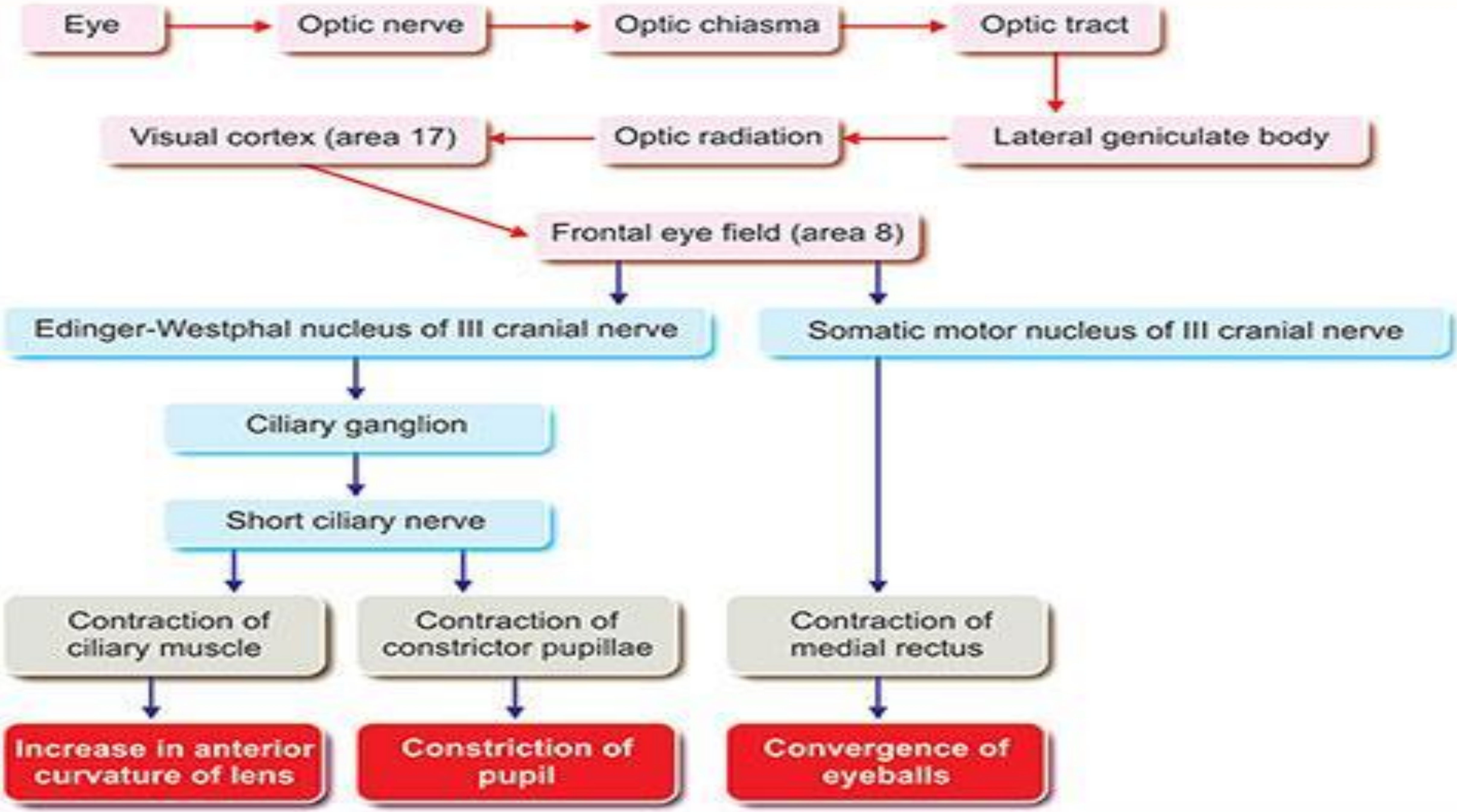
- Visual impulses from retina travel through the ***optic nerve***, ***optic chiasma***, ***optic tract***, ***lateral geniculate body*** & ***optic radiation*** to the visual cortex(***area 17***) of occipital lobe.
- the associating fibers then carry the afferent impulses from visual cortex(area 17) to the frontal lobe of cerebral cortex.

EFFERENT PATHWAY

- From ***area 8*** of cerebral cortex, the fibers pass to the caudal portion of ***Edinger-Westphal nucleus***.
- The ***preganglionic fibers*** then pass through the ***CN III*** to the ***ciliary ganglion***.
- From ciliary ganglion, ***postganglionic fibers*** pass through the ***short ciliary nerves*** & supply the ***ciliary muscle*** & ***constrictor pupillae muscle***.

CENTER FOR ACCOMODATION REFLEX

- The center for accommodation reflex lies in the frontal eye field called ***area 8*** which is situated in the ***frontal lobe of cerebral cortex***.


Eye
Optic nerve
Optic chiasma
Optic tract
Lateral geniculate body
Optic radiation
Visual cortex (area 17)
Frontal eye field (area 8)
Edinger-Westphal nucleus of III cranial nerve
Somatic motor nucleus of III cranial nerve
Ciliary ganglion
Short ciliary nerve
Contraction of ciliary muscle
Contraction of constrictor pupillae
Contraction of medial rectus
Increase in anterior curvature of lens
Constriction of pupil
Convergence of eyeballs

RANGE OF ACCOMODATION

FAR POINT(PUNCTUM REMOTUM)

- The farthest point from the eye at which the object can be seen is called far point or punctum remotum.
- In normal eye far point is infinite i.e. distance beyond 6 meters or 20 feet.

NEAR POINT(PUNCTUM PROXIMUM)

- The nearest point from eye at which the object is seen clearly is called near point or punctum proximum.
- The near point for an eye is between 7 to 40cm depending upon the age.

Distance between far point & near point is called as range of accommodation.

AMPLITUDE OF ACCOMODATION

STATIC REFRACTION

- The refractive power of eye during far vision is called static refraction.
- It is represented by ‘**R**’.

DYNAMIC REFRACTION

- The refractive power of eye during near vision is called as dynamic refraction.
- It is represented by ‘**P**’.

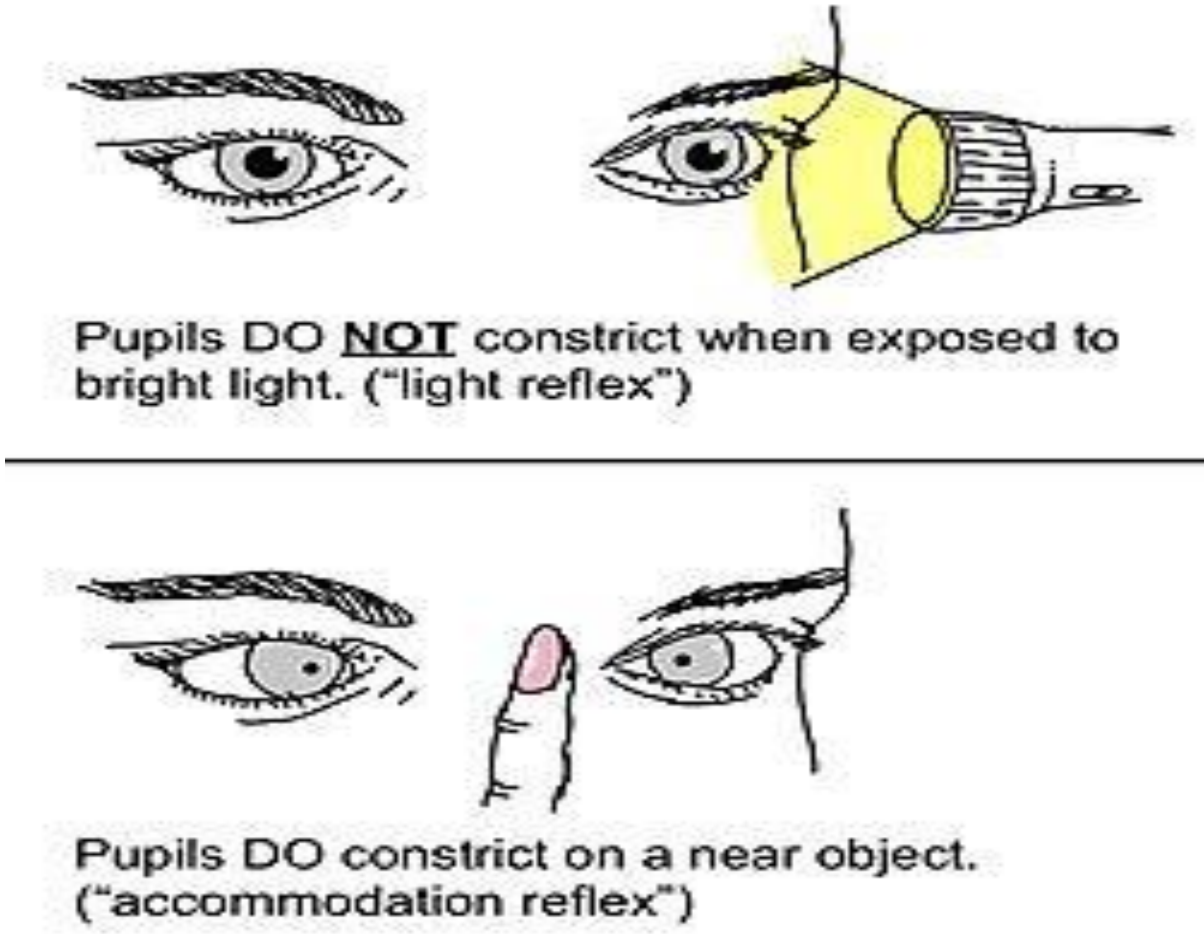
The difference between the dynamic refraction & the static refraction(P-R) is called as amplitude of refraction.

- It is expressed as ***diopter(D)***.

APPLIED PHYSIOLOGY

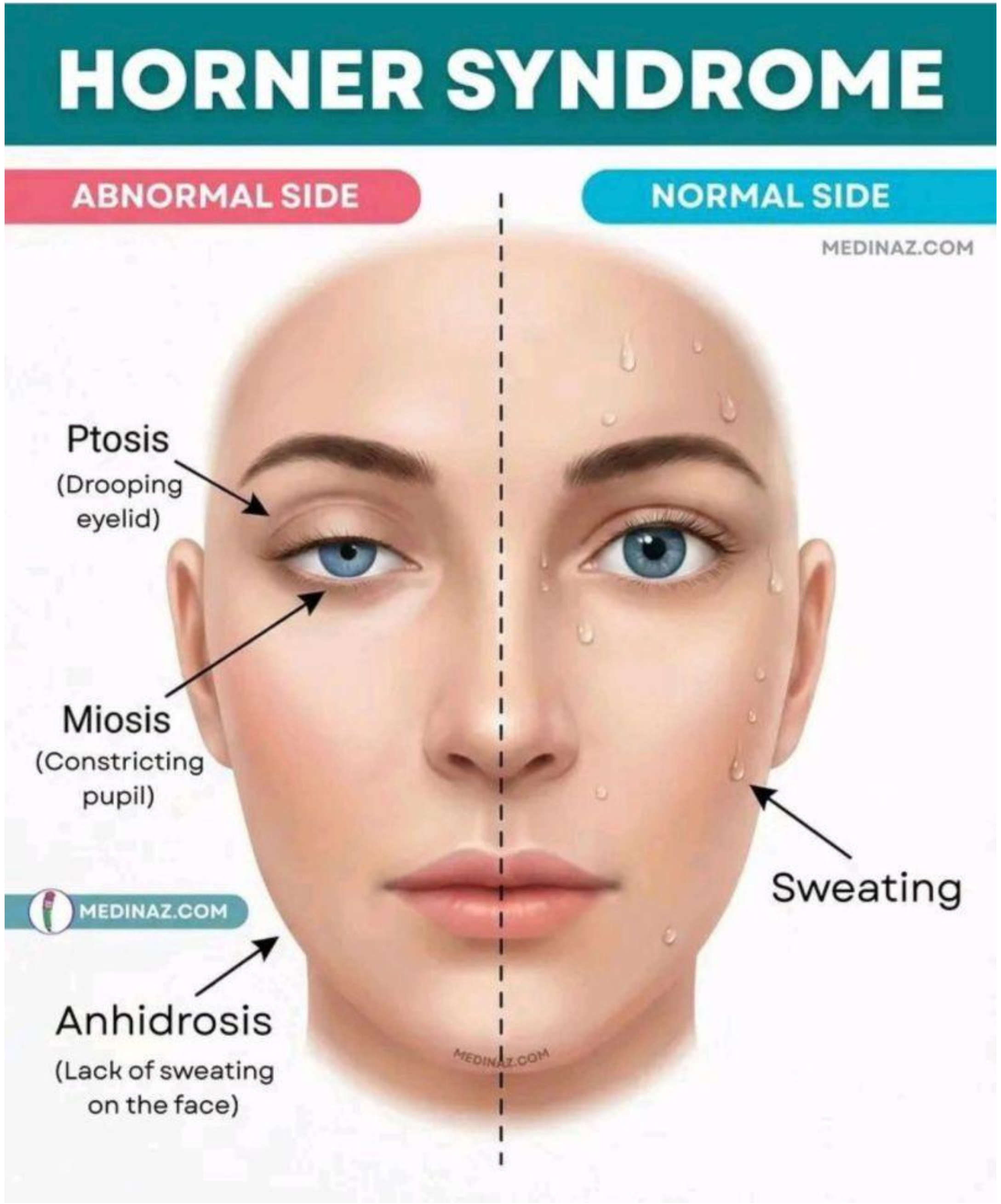
ARGYLL ROBERTSON PUPIL

- Clinical condition in which light reflex of a person is lost but the accommodation reflex is normal.
- It occurs in tertiary syphilis, diabetic neuropathy, alcoholic neuropathy.



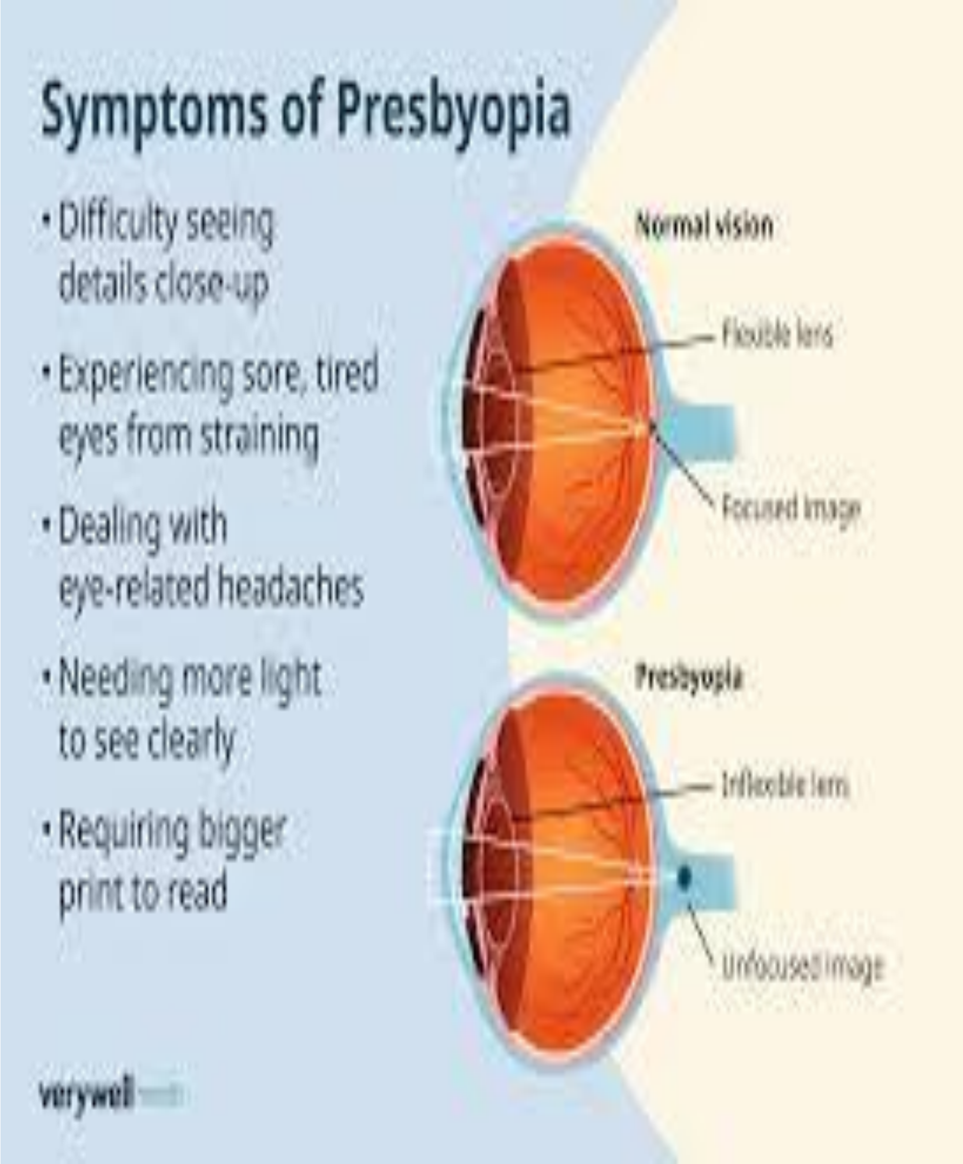
HORNER SYNDROME

- Disorder characterized by drooping of upper eyelid(ptosis), swelling of lower eyelid, abnormal pupil constriction(myosis), sinking of eyeball into its cavity(enopthalmos) & absence of sweating on the affected side of face.



PRESBYOPIA

- In old age amplitude of accommodation is decreased & near point is away from the eye, this condition is termed as presbyopia.



POST ASSESSMENT

Q1. The parasympathetic fibers controlling pupillary constriction synapse in the:

a) Superior cervical ganglion
b) Trigeminal ganglion
c) Ciliary ganglion
d) Otic ganglion
e) Pterygopalatine ganglion

Q2. The efferent limb of pupillary light reflex is carried by:

a) CN I
b) CN II
c) CN III
d) CN IV
e) CN V

Cont..

Q3. Accomodation increases the refractive power of eye by making the lens:

a) Flatter
b) More convex
c) Smaller
d) More concave
e) Less elastic

Q4. Preganglionic parasympathetic neurons involved in accommodation originate in the:

a) Superior cervical ganglion
b) Lateral geniculate body
c) Superior colliculus
d) Trigeminal nucleus
e) Edinger-westphal nucleus